

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

CONTENTS

1. Introduction	1
2. Preliminaries	2
3. Analytic tools for Weyl chamber geometry	6
4. Cwikel's estimate in $N > 2$	13
5. Better Cwikel estimate	15
References	26

1. INTRODUCTION

Definition 1.1. Suppose $1 < p < \infty$. We say that a function $f \in L^p_{\text{loc}}(\mathbb{R}^N, dw)$ belongs to Dunkl Besov space $B^p_{\text{Dunkl}}(\mathbb{R}^N)$ if

$$\|f\|_{B^p_{\text{Dunkl}}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left(\frac{d(x, y)}{\|x - y\|} \right)^p \frac{|f(x) - f(y)|^p}{w(B(x, d(x, y)))^2} dw(x) dw(y) \right)^{1/p} < +\infty.$$

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ -measurable set E by

$$(1.1) \quad (f)_E := \int_E f(x) d\mu(x) := \frac{1}{\mu(E)} \int_E f(x) d\mu(x).$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

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Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper. We will denote by $O(x) = \{\sigma(x) : \sigma \in G\}$ the G -orbit of a point $x \in \mathbb{R}^N$, and $O(E) = \bigcup_{x \in E} O(x)$ the G -orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2.2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$ as the distance between two G -orbits $O(x)$ and $O(y)$. Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber Γ .

It is clear that $O(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(O(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = O(B(x, r))$.

Denote by $K_{\text{Dunkl},j}(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \dots, N\}$ and every x, y with $d(x, y) \neq 0$,*

$$|K_{\text{Dunkl},j}(x, y)| \leq C \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

2.2. Systems of dyadic cubes.

Definition 2.3. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathcal{D} := \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$, $\mathcal{D}_k := \{Q_\alpha^k : \alpha \in \mathcal{A}_k\}$, of Borel sets $Q_\alpha^k \subseteq X$ is called a *system of dyadic cubes over X* if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha \in \mathcal{A}_k} Q_\alpha^k;$$

(ii) If $\ell \geq k$, then either $Q_\beta^\ell \subseteq Q_\alpha^k$ or $Q_\alpha^k \cap Q_\beta^\ell = \emptyset$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q_\alpha^k \subseteq Q_\beta^\ell$;

(iv) For each (k, α) there exist 2^N indices β such that

$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q \in \mathcal{D}_{k+1} \\ Q \subseteq Q_\alpha^k}} Q;$$

(v) For each (k, α) , one has

$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$

for some positive constants A_1, A_2 depending only on the geometry of X , where we set $B_X(x, r) := B(x, r) \cap X$ for all $x \in X$ and $r > 0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then

$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a *dyadic cube of generation k* , with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.4. For every dyadic cube $Q \in \mathcal{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.5. Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with $(m_1, \dots, m_N) \in \mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber Γ , whose construction will be given in Corollary 3.7.

Lemma 2.6. There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k \in \mathbb{Z}$, there exists a cube $Q \in \bigcup_{v=1}^{2^N} \mathbb{D}_k^v$ such that

$$B \subset Q \subset 12 \sqrt{N} B.$$

Proof. • **Step 1.** Recall the one-dimensional setting given in [10, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.5). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \left\{ \left\lfloor \frac{m}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j}, \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\rfloor \mid m \in \mathbb{Z} \right\}.$$

These choices together with the nestedness property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

- (1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
- (2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
- (3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a - b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J . Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• **Step 2.** We extend the construction to $\mathbb{R}^N$ ($N \geq 1$).

For $\vartheta = (\vartheta_1, \vartheta_2, \dots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^N I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i -th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \dots, \vartheta_N)$, and let $Q = \prod_{j=1}^N I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N}\ell(Q) \leq 12\sqrt{N}r_B.$$

This implies that $Q \subset B(x_B, 12\sqrt{N}r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \dots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.6. $\square$

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [2–4, 8, 9, 11–14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Definition 2.7. Let $\mathcal{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q \in \mathcal{D}}$ is an NWO sequence if

- (1) e_Q is supported in cQ for some positive constant $c \in 2\mathbb{N} + 1$ independent of Q ;
- (2) $\|e_Q\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq w(Q)^{\frac{1}{r}-\frac{1}{2}}$ for some $r > 2$;

Lemma 2.8. Let $0 < p < \infty$, $0 < q \leq \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that

$$A = \sum_{Q \in \mathcal{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathcal{Q}}$ and $\{f_Q\}_{Q \in \mathcal{Q}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathcal{Q}}$ is a sequence of scalars. We have

$$(2.5) \quad \|A\|_{S^{p,q}} \leq C_{N,p,q,w} \|\{\lambda_Q\}_{Q \in \mathcal{Q}}\|_{\ell^{p,q}}.$$

Proof. This property can be found in [12, (1.9)] (see also [8, Section 7]). $\square$

3. ANALYTIC TOOLS FOR WEYL CHAMBER GEOMETRY

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \text{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [6, Definition 2.7] A connected component of

$$(3.1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

Definition 3.2. [6, Definition 2.9] A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. With the notation given in Definition 3.2, Δ is a basis of W . If we denote by $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$, then the open and closed chambers can be written as

$$(3.2) \quad \Gamma_+ = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\},$$

$$(3.3) \quad \Gamma = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\},$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+ . By [6, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha \rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \text{span}(\Delta)$ and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion $\text{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\text{span}(\Delta) = W$. On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W , and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [6, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+ , note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ , specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. $\square$

Notation 3.4. *With the notation given in Lemma 3.3, denote by*

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ .

Lemma 3.5. *Each Weyl chamber Γ admits the decomposition*

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. $\square$

Corollary 3.6. *Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let*

$$\begin{aligned} \Phi_W : W &\longrightarrow \mathbb{R}^{N_0}, & \Phi_W(w) &:= (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle), \\ \Phi : \mathbb{R}^N = W \oplus W^\perp &\longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, & \Phi(w + z) &:= (\Phi_W(w), \Phi_{W^\perp}(z)). \end{aligned}$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since Δ is a basis of W , the Gram matrix $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \longrightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. $\square$

Corollary 3.7. *Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ .*

Proof. Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}_{\Omega,k}^0$ to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Set

$$\mathbb{D}_{\Omega}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that $\mathbb{D}_{\Omega}^0$ is a system of dyadic cubes over Ω .

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_{\Gamma} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_{\Gamma}$ is a system of dyadic cubes over Γ . $\square$

Notation 3.8. Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define

$$(3.4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

Lemma 3.9. We have

$$(3.5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each $v \in R_+$, the coefficients $a_{v,i} \geq 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$ and satisfy $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω ,

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) dw(x).$$

We make the linear change of variables $(u, z) = \Phi(x)$, so that $x = \Phi^{-1}(u, z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i \rangle$ for $i = 1, \dots, N_0$. This implies that

$$(3.6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) \left(\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx$$

$$= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz.$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. $\square$

Lemma 3.10. *Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and*

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
- (2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2 . By [6, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

- **Step 1.** *Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.*

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1} \sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

- **Step 2.** *Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.*

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \left\{ x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0 \right\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. $\square$

Definition 3.11. We say that $R' \subset \Omega$ is an axis-parallel rectangle if

$$R' := I_1 \times \dots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$(3.7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_\Omega(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\text{dbl}, \Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_\Omega(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_\Omega(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2 \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on L_* and m_* , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We *claim* that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0} \theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c . Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \leq \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4 \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. $\square$

Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$(3.8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$. It follows from (3.8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$(3.9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 3.12, yields

$$(3.10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Lemma 3.13. *For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis-parallel rectangle R' on Ω such that*

$$(3.11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(3.12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis-parallel rectangle

$$(3.13) \quad R' := \left(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. $\square$

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. *There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,*

$$(3.14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

In particular, there exists a constant $C_{\text{dbl},\Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl},\Gamma} w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$(3.15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$(3.16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying μ to the inclusions (3.9) and using (3.16) yields

$$(3.17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (3.17) with Lemma 3.12 and the equality (3.6), we obtain that

$$(3.18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (3.14). The second statement follows immediately from (3.14). The third statement follows from the inequalities (3.14) and (2.1). This completes the proof of Theorem 3.14. $\square$

4. CBIKEL'S ESTIMATE IN $N > 2$

Lemma 4.1. *Assume that $N > 1$. Then for every pair of multi-indices α, β , there exists a constant $C_{\alpha,\beta} > 0$ such that*

$$(4.19) \quad \left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \right| \leq C_{\alpha,\beta} \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))}$$

for all $x, y \in \mathbb{R}^N$ satisfying $d(x, y) > 0$.

Proof. Let $h_t(x, y)$ denote the integral kernel of the Dunkl heat semigroup $e^{t\Delta_{\text{Dunkl}}}$. By [1, Theorem 4.1(d)], for every pair of multi-indices α, β , there exist constants $C_{\alpha,\beta}, c_{\alpha,\beta} > 0$ such that

$$(4.20) \quad \left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \leq C_{\alpha,\beta} t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c_{\alpha,\beta} \frac{d(x, y)^2}{t}\right)$$

for all $t > 0$ and $x, y \in \mathbb{R}^N$.

By the spectral functional calculus,

$$(-\Delta_{\text{Dunkl}})^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{t\Delta_{\text{Dunkl}}} \frac{dt}{\sqrt{t}}.$$

Consequently, for $d(x, y) > 0$,

$$(4.21) \quad K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty h_t(x, y) \frac{dt}{\sqrt{t}}.$$

It follows that $K_{(-\Delta_{\text{Dunkl}})^{-1/2}}$ is smooth whenever $d(x, y) > 0$ and

$$(4.22) \quad \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty \partial_x^\alpha \partial_y^\beta h_t(x, y) \frac{dt}{\sqrt{t}}.$$

Set $m := |\alpha| + |\beta|$. Fix $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$. By (4.20) and (4.22),

$$(4.23) \quad \left| \partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y) \right| \lesssim \int_0^\infty t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt.$$

Suppose first that $0 < t \leq d(x, y)^2$. Then $\sqrt{t} \leq d(x, y)$. By the upper estimate in (2.2), for $z = x, y$,

$$\frac{w(B(z, d(x, y)))}{w(B(z, \sqrt{t}))} \lesssim \left(\frac{d(x, y)}{\sqrt{t}} \right)^N.$$

Equivalently,

$$w(B(z, \sqrt{t})) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N w(B(z, d(x, y))).$$

Taking the maximum over $z = x, y$ gives

$$(4.24) \quad V(x, y, \sqrt{t}) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N V(x, y, d(x, y)).$$

Therefore,

$$\begin{aligned} & \int_0^{d(x, y)^2} t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \\ & \lesssim \frac{d(x, y)^N}{V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+m+1}{2}} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt. \end{aligned}$$

Using again the change of variables $s = \frac{d(x, y)^2}{t}$, we obtain

$$\int_0^{d(x, y)^2} t^{-\frac{N+m+1}{2}} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt = d(x, y)^{1-N-m} \int_1^\infty s^{\frac{N+m-3}{2}} e^{-cs} ds \lesssim d(x, y)^{1-N-m}.$$

Consequently,

$$(4.25) \quad \int_0^{d(x, y)^2} t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Suppose next that $t \geq d(x, y)^2$. Then $\sqrt{t} \geq d(x, y)$. By the lower estimate in (2.2), for $z = x, y$,

$$\frac{w(B(z, \sqrt{t}))}{w(B(z, d(x, y)))} \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N.$$

It follows that

$$(4.26) \quad V(x, y, \sqrt{t}) \gtrsim \left(\frac{\sqrt{t}}{d(x, y)} \right)^N V(x, y, d(x, y)).$$

Hence

$$\int_{d(x, y)^2}^\infty t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^N}{V(x, y, d(x, y))} \int_{d(x, y)^2}^\infty t^{-\frac{N+m+1}{2}} dt.$$

Since $N > 1$ and $m \geq 0$, we have $N + m > 1$, and therefore

$$\int_{d(x, y)^2}^\infty t^{-\frac{N+m+1}{2}} dt = \frac{2}{N + m - 1} d(x, y)^{1-N-m}.$$

Thus

$$(4.27) \quad \int_{d(x,y)^2}^{\infty} t^{-\frac{m+1}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Combining (4.23), (4.25), and (4.27), we obtain

$$|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)| \lesssim \frac{d(x, y)^{1-m}}{V(x, y, d(x, y))}.$$

Since $m = |\alpha| + |\beta|$, this gives

$$|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)| \lesssim \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))},$$

which is exactly (4.19). $\square$

Lemma 4.2. *Assume that $N > 2$. There exists a constant $C_{N,R,k} > 0$ such that, for every $f \in L_N(\mathbb{R}^N, dx)$,*

$$M_f(-\Delta_{\text{Dunkl}})^{-\frac{1}{2}} \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$$

and

$$(4.28) \quad \|M_f(-\Delta_{\text{Dunkl}})^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq C_{N,R,k} \|f\|_{L_N(\mathbb{R}^N, dx)}.$$

Proof. For $g \in G$, let $U_g : \xi \rightarrow \xi \circ g^{-1}$. For a chamber Γ and $g \in G$, define an integral operator

$$V_{\Gamma,g} = M_{\chi_\Gamma}(-\Delta_{\text{Dunkl}})^{-\frac{1}{2}} M_{\chi_{g\Gamma}} U_g.$$

By Lemma 4.1, each operator $V_{\Gamma,g}$ satisfies the assumptions in Theorem 5.1. We, obviously, have

$$M_f(-\Delta_{\text{Dunkl}})^{-\frac{1}{2}} = \sum_{\Gamma} \sum_{g \in G} M_f V_{\Gamma,g} \cdot U_{g^{-1}}.$$

The assertion follows now from Theorem 5.1. $\square$

5. BETTER CBIKEL ESTIMATE

Theorem 5.1. *Let Γ be a chamber. Let V be an integral operator on $L_2(\Gamma, w(x)dx)$ with integral kernel K satisfying the conditions*

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \leq C_{\alpha,\beta} \frac{|x - y|^{1-|\alpha|-|\beta|}}{V(x, |x - y|)}.$$

We have

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(\Gamma, w(x)dx)} \leq c_K \|f\|_{L_N(\Gamma)}.$$

Notation 5.2. *Fix a closed Weyl chamber Γ and let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism appearing in Corollary 3.6.*

Let $\mathbb{D}_\Gamma$ be the dyadic system over Γ constructed in Corollary 3.7. If $I \in \mathbb{D}_\Gamma$ and if $Q = \Phi(I)$, then we denote

$$d_I = \ell(Q).$$

For every $I \in \mathbb{D}_\Gamma$ and for every $f \in L_r^{\text{loc}}(\mathbb{R}^N, w(x)dx)$, define

$$(5.29) \quad \lambda_I(f) = d_I w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)}.$$

Lemma 5.3. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$(5.30) \quad \text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(5.31) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every $x \in \Gamma$, there exists a unique $I \in \mathbb{D}_{\Gamma,j}$ containing x . For this I , we have $d_I = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(\Phi(x) - \Phi(y))) = \sum_{j \in \mathbb{Z}} \left(\sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \right) \psi(2^j(\Phi(x) - \Phi(y))) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

If $x \in I$ and $\psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) \neq 0$, then

$$\Phi(x) \in \Phi(I) = Q, \quad |\Phi(x) - \Phi(y)| \leq d_I = \ell(Q).$$

We obviously have $|\Phi(x) - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|\Phi(x) - \Phi(y)|_\infty \leq \ell(Q)$. Thus, $|\Phi(y) - c_Q| \leq \frac{3}{2}\ell(Q)$. In other words, $\Phi(y) \in 3Q$ or, equivalently, $y \in 3I$. Thus,

$$\chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

This completes the proof. $\square$

Lemma 5.4. *Assume that $N > 1$ and $2 < r < \infty$. Let $f \in L_r^{\text{loc}}(\mathbb{R}^N, w(x)dx)$. There exist functions $\{e_{k,l,I}\}_{k,l \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}$, $\{f_{k,l,I}\}_{k,l \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}$, such that*

(1) $e_{k,l,I}$ is supported in I and $f_{k,l,I}$ is supported in $3I$;

(2)

$$\|e_{k,l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \|f\|_{L_r(\mathbb{R}^N, w(x)dx)}, \quad \|f_{k,l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq c_{K,\kappa} d_I w(I)^{\frac{1}{r}-1};$$

(3) For almost every $x, y \in \Gamma$, we have

$$(5.32) \quad f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-2N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k,l,I}(x) f_{k,l,I}(y).$$

Proof. We have

$$\begin{aligned} K(x, y) &= \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \\ &= \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) + \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \not\subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

We will only consider the first sum. The second one can be treated similarly, but the notations become heavier. **feel free to write the expression for the second sum if you think it is needed. the only difference is that the domain of the map K_I becomes more complicated.**

For every $I \in \mathbb{D}_\Gamma$ with $3I \subset \Gamma$, define the map

$$K_I : [-\frac{1}{2}, \frac{1}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N \rightarrow \mathbb{C}$$

by setting

$$K_I(a, b) = K(c_I + d_I \Phi^{-1}(a), c_I + d_I \Phi^{-1}(b)) \psi(a - b), \quad a \in [-\frac{1}{2}, \frac{1}{2}]^N, \quad b \in [-\frac{3}{2}, \frac{3}{2}]^N.$$

It follows from the assumption on K that **feel free to prove this if you think this requires a proof. however, do not prove it here — make a separate lemma instead.**

$$\|K_I\|_{C^{4N+2}([- \frac{1}{2}, \frac{1}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq c_K \frac{d_I}{w(I)}.$$

Hence, there exists an extension $K_I : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_I\|_{C^{4N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq c'_K \frac{d_I}{w(I)}.$$

In particular,

$$\|K_I\|_{W^{4N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq c''_K \frac{d_I}{w(I)}.$$

We may, therefore, write

$$K_I(a, b) = \sum_{k, l \in \mathbb{Z}^N} \text{coef}_{k, l, I} e^{i\langle k, a \rangle} e^{i\langle l, b \rangle}, \quad a \in [-\frac{1}{2}, \frac{1}{2}]^N, \quad b \in [-\frac{3}{2}, \frac{3}{2}]^N,$$

where

$$\sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{4N+2} |\text{coef}_{k, l, I}|^2 \leq (c''_K \frac{d_I}{w(I)})^2.$$

In particular,

$$|\text{coef}_{k, l, I}| \leq c''_K (1 + |k|^2 + |l|^2)^{-2N-1} \frac{d_I}{w(I)}.$$

This allows us to write

$$\begin{aligned} \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) &= \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) K_I\left(\frac{\Phi^{-1}(x - c_I)}{d_I}, \frac{\Phi^{-1}(y - c_I)}{d_I}\right) = \\ &= \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) \sum_{k, l \in \mathbb{Z}^N} \text{coef}_{k, l, I} e^{i\langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle} e^{i\langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle} = \\ &= \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-2N-1} \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} (1 + |k|^2 + |l|^2)^{2N+1} \text{coef}_{k, l, I} \cdot \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle} \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}. \end{aligned}$$

We now set

$$\begin{aligned} e_{k, l, I}(x) &= f(x) \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle}, \\ f_{k, l, I}(y) &= (1 + |k|^2 + |l|^2)^{2N+1} \text{coef}_{k, l, I} \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}. \end{aligned}$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,I}) &\subset I, \quad \text{subset}(f_{k,l,I}) \subset 3I, \\ \|e_{k,l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \|f\|_{L_r(I, w(x)dx)}, \\ \|f_{k,l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= (1 + |k|^2 + |l|^2)^{2N+1} |\text{coef}_{I,k,l}| \cdot w(3I)^{\frac{1}{r}} \leq c_K'' \frac{d_I}{w(I)} \cdot w(3I)^{\frac{1}{r}} \leq c_{K,\kappa} d_I w(I)^{\frac{1}{r}-1}. \end{aligned}$$

□

Lemma 5.5. *We have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{K,\kappa} \left\| \left\{ d_I w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{l_{N,\infty}}.$$

Proof. The assertion follows from Lemma 5.4, Lemma 2.8 and the triangle inequality. **feel free to expand the argument if needed.** □

Lemma 5.6. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. For every $Q \in \mathbb{D}^0$, set*

$$b_Q := |Q|^{\frac{1}{s}-1} \int_Q |F(x)| dx = |Q|^{-1/s'} \int_Q |F(x)| dx,$$

Then

$$(5.33) \quad \#\{Q \in \mathbb{D}^0 : b_Q > t\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

Equivalently,

$$\left\| \{b_Q\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be an arbitrary finite collection of dyadic cubes, and define

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

We first claim that

$$(5.34) \quad \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

Indeed, fix

$$x \in \bigcup_{Q \in \mathcal{E}} Q.$$

Set

$$\mathcal{E}_x := \{Q \in \mathcal{E} : x \in Q\}.$$

Since $\mathcal{E}$ is finite, there exists a cube $Q_x \in \mathcal{E}_x$ having minimal volume. Since any two cubes in the same dyadic grid which contain the same point are nested, every $Q \in \mathcal{E}_x$ contains Q_x . Thus the cubes in $\mathcal{E}_x$ belong to the chain of dyadic ancestors of Q_x .

Denote by $Q_x^{(k)}$ the k -th dyadic ancestor of Q_x , so that

$$|Q_x^{(k)}| = 2^{kN} |Q_x|, \quad k \geq 0.$$

It follows that

$$G_{\mathcal{E}}(x) = \sum_{Q \in \mathcal{E}_x} |Q|^{-1/s'}$$

$$\begin{aligned}
&\leq \sum_{k=0}^{\infty} |Q_x^{(k)}|^{-1/s'} \\
&= |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \\
&\lesssim_{N,s} |Q_x|^{-1/s'}.
\end{aligned}$$

For each $Q \in \mathcal{E}$, define

$$E_Q := \left\{ x \in \bigcup_{P \in \mathcal{E}} P : Q_x = Q \right\}.$$

The sets $\{E_Q\}_{Q \in \mathcal{E}}$ are pairwise disjoint and satisfy

$$E_Q \subset Q.$$

Therefore,

$$\begin{aligned}
\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} &= \int_{\mathbb{R}^N} G_{\mathcal{E}}(x)^{s'} dx \\
&\lesssim_{N,s} \sum_{Q \in \mathcal{E}} \int_{E_Q} |Q|^{-1} dx \\
&= \sum_{Q \in \mathcal{E}} \frac{|E_Q|}{|Q|} \\
&\leq \sum_{Q \in \mathcal{E}} 1 = \#\mathcal{E}.
\end{aligned}$$

Taking the s' -th root proves (5.34).

We now estimate the sum of $\{b_Q\}_{Q \in \mathcal{E}}$. By Hölder's inequality and (5.34),

$$\begin{aligned}
\sum_{Q \in \mathcal{E}} b_Q &= \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \\
&= \int_{\mathbb{R}^N} |F(x)| \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x) dx \\
&= \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \\
&\leq \|F\|_{L^s(\mathbb{R}^N)} \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \\
&\lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.
\end{aligned}$$

Now fix $t > 0$ and let

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\}$$

be an arbitrary finite subcollection. Since $b_Q > t$ for every $Q \in \mathcal{E}$, we have

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} b_Q \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

If $\mathcal{E} \neq \emptyset$, dividing by $(\#\mathcal{E})^{1/s'}$ and using

$$1 - \frac{1}{s'} = \frac{1}{s},$$

we obtain

$$(\#\mathcal{E})^{1/s} \lesssim_{N,s} t^{-1} \|F\|_{L^s(\mathbb{R}^N)}.$$

Consequently,

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since this estimate holds for every finite subcollection

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\},$$

it follows that

$$\#\{Q \in \mathbb{D}^0 : b_Q > t\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

This proves (5.33), and hence also the stated weak ℓ^s estimate. $\square$

Lemma 5.7. *Assume that $N > 2$ and $2 < r < N$. Then, for every $\sigma \in G$ and every $f \in L^N(\mathbb{R}^N, dx)$, we have $f \in L_{\text{loc}}^r(\mathbb{R}^N, dw)$ and*

$$(5.35) \quad \left\| \{\lambda_{\sigma,I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

The implicit constant is independent of σ and f .

Proof. We first note that $f \in L_{\text{loc}}^r(\mathbb{R}^N, dw)$. Indeed, the density of dw with respect to Lebesgue measure is $\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)}$, which is bounded on every compact subset of $\mathbb{R}^N$. Hence, if $E \subset \mathbb{R}^N$ is bounded, then, since $r < N$,

$$\int_E |f(x)|^r dw(x) \lesssim_E \int_E |f(x)|^r dx \leq |E|^{1-\frac{r}{N}} \|f\|_{L^N(\mathbb{R}^N, dx)}^r < \infty.$$

Thus $\lambda_{\sigma,I}(f)$ in (5.29) is well defined.

Fix $\sigma \in G$. Recall that $\Omega = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$ and, by Lemma 3.9,

$$(5.36) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)},$$

where $a_{v,i} \geq 0$. In particular, ϕ is nondecreasing in each of the variables $u_1, \dots, u_{N_0}$ and is independent of the remaining $N - N_0$ variables.

Let $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$, where

$$Q = \prod_{i=1}^N [m_i \ell, (m_i + 1)\ell) \subset \Omega, \quad \ell = \ell(Q) = \ell(I),$$

with

$$m_i \in \mathbb{Z}_+, \quad 1 \leq i \leq N_0,$$

and

$$m_i \in \mathbb{Z}, \quad N_0 < i \leq N.$$

Define $u^+ := ((m_1 + 1)\ell, \dots, (m_{N_0} + 1)\ell)$ and

$$Q^+ := \prod_{i=1}^{N_0} \left[\left(m_i + \frac{1}{2} \right) \ell, (m_i + 1)\ell \right) \times \prod_{i=N_0+1}^N [m_i \ell, (m_i + 1)\ell).$$

Then

$$|Q^+| = 2^{-N_0}|Q|.$$

For every $(u, z) \in Q^+$ and every $1 \leq i \leq N_0$,

$$u_i \geq \left(m_i + \frac{1}{2}\right)\ell \geq \frac{1}{2}(m_i + 1)\ell.$$

Since $a_{v,i} \geq 0$, it follows that, for every $v \in R_+$,

$$\sum_{i=1}^{N_0} a_{v,i} u_i \geq \frac{1}{2} \sum_{i=1}^{N_0} a_{v,i} (m_i + 1)\ell.$$

Consequently,

$$\phi(u) \geq 2^{-2 \sum_{v \in R_+} k(v)} \phi(u^+).$$

Since

$$2 \sum_{v \in R_+} k(v) = \sum_{v \in R} k(v) = \mathbf{N} - N,$$

we obtain

$$\phi(u) \geq 2^{-(\mathbf{N}-N)} \phi(u^+), \quad (u, z) \in Q^+.$$

Therefore,

$$\begin{aligned} \mu(Q) &= \int_Q \phi(u) \, du \, dz \\ &\geq \int_{Q^+} \phi(u) \, du \, dz \\ &\geq 2^{-(\mathbf{N}-N)} |Q^+| \phi(u^+) \\ &= 2^{-(\mathbf{N}-N+N_0)} |Q| \phi(u^+). \end{aligned}$$

On the other hand, by the coordinatewise monotonicity of ϕ ,

$$\operatorname{ess\,sup}_{(u,z) \in Q} \phi(u) \leq \phi(u^+).$$

It follows that

$$(5.37) \quad \operatorname{ess\,sup}_Q \phi \lesssim \frac{\mu(Q)}{|Q|}.$$

Since dw is G -invariant and

$$\mu(Q) = w(\Phi^{-1}(Q)) = w(I),$$

we have

$$\begin{aligned} \frac{1}{w(I)} \int_{\sigma I} |f(x)|^r \, dw(x) &= \frac{1}{w(I)} \int_I |f(\sigma x)|^r \, dw(x) \\ &= \frac{1}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r \, d\mu(\xi) \\ &= \frac{1}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r \phi(\xi) \, d\xi \\ &\leq \frac{\operatorname{ess\,sup}_Q \phi}{\mu(Q)} \int_Q |f(\sigma \Phi^{-1}(\xi))|^r \, d\xi \end{aligned}$$

$$\lesssim \frac{1}{|Q|} \int_Q |f(\sigma\Phi^{-1}(\xi))|^r d\xi,$$

where we used (5.37) in the last inequality. Hence, by the definition (5.29),

$$\begin{aligned} \lambda_{\sigma,I}(f)^r &= \ell(I)^r \frac{1}{w(I)} \int_{\sigma I} |f(x)|^r dw(x) \\ (5.38) \quad &\lesssim \frac{\ell(Q)^r}{|Q|} \int_Q |f(\sigma\Phi^{-1}(\xi))|^r d\xi. \end{aligned}$$

Set

$$s := \frac{N}{r}.$$

Since $r < N$, we have $s > 1$. Moreover,

$$|Q| = \ell(Q)^N,$$

and hence

$$\frac{\ell(Q)^r}{|Q|} = |Q|^{\frac{r}{N}-1} = |Q|^{\frac{1}{s}-1}.$$

Define

$$F_\sigma(\xi) := \begin{cases} |f(\sigma\Phi^{-1}(\xi))|^r, & \xi \in \Omega, \\ 0, & \xi \notin \Omega. \end{cases}$$

Then (5.38) gives

$$(5.39) \quad \lambda_{\sigma,I}(f)^r \lesssim |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi.$$

We next verify that $F_\sigma \in L^s(\mathbb{R}^N)$. Since $rs = N$, a change of variables $\xi = \Phi(x)$ gives

$$\begin{aligned} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s &= \int_\Omega |f(\sigma\Phi^{-1}(\xi))|^{rs} d\xi \\ &= \int_\Omega |f(\sigma\Phi^{-1}(\xi))|^N d\xi \\ &= |\det(\Phi)| \int_\Gamma |f(\sigma x)|^N dx \\ &= |\det(\Phi)| \int_{\sigma\Gamma} |f(x)|^N dx \\ &\leq |\det(\Phi)| \|f\|_{L^N(\mathbb{R}^N, dx)}^N < \infty, \end{aligned}$$

where in the fourth line we used the fact that σ is orthogonal. Thus $F_\sigma \in L^s(\mathbb{R}^N)$.

Since $\mathbb{D}_\Omega^0$ is a subfamily of the standard dyadic system $\mathbb{D}^0$, we may apply Lemma 5.6 to F_σ . Together with (5.39), this gives, for every $a > 0$,

$$\begin{aligned} \#\{I \in \mathbb{D}_\Gamma : \lambda_{\sigma,I}(f) > a\} &\leq \#\left\{Q \in \mathbb{D}_\Omega^0 : |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi \gtrsim a^r\right\} \\ &\leq \#\left\{Q \in \mathbb{D}^0 : |Q|^{\frac{1}{s}-1} \int_Q F_\sigma(\xi) d\xi \gtrsim a^r\right\} \\ &\lesssim (a^r)^{-s} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s \\ &= a^{-rs} \|F_\sigma\|_{L^s(\mathbb{R}^N)}^s \end{aligned}$$

$$\lesssim a^{-N} \|f\|_{L^N(\mathbb{R}^N, dx)}^N,$$

where in the last line we used $rs = N$ and the estimate for $\|F_\sigma\|_{L^s(\mathbb{R}^N)}^s$ obtained above. Therefore,

$$\#\{I \in \mathbb{D}_\Gamma : \lambda_{\sigma, I}(f) > a\} \lesssim \left(\frac{\|f\|_{L^N(\mathbb{R}^N, dx)}}{a} \right)^N, \quad a > 0.$$

By the definition of the weak ℓ^N quasi-norm, this is exactly

$$\left\| \{\lambda_{\sigma, I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

All the implicit constants above depend only on the fixed structural parameters and are independent of $\sigma \in G$ and f . This completes the proof. $\square$

Lemma 5.8. *Assume that $N > 2$ and $2 < r < N$. Let*

$$\left\{ e_{p, q, I}^{\sigma, \tau} \right\}_{p, q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G} \quad \text{and} \quad \left\{ h_{p, q, I}^{\sigma, \tau} \right\}_{p, q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma, \sigma, \tau \in G}$$

be the families obtained from Lemma 5.4 with this choice of r . For every

$$f \in L^N(\mathbb{R}^N, dx), \quad \sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N,$$

define

$$A_{\sigma, \tau, p, q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma, I}(f) \left\langle \cdot, \overline{h_{p, q, I}^{\sigma, \tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} e_{p, q, I}^{\sigma, \tau}.$$

Then

$$A_{\sigma, \tau, p, q}(f) \in S^{N, \infty}(L^2(\mathbb{R}^N, dw))$$

and

$$(5.40) \quad \|A_{\sigma, \tau, p, q}(f)\|_{S^{N, \infty}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)},$$

where the implicit constant is independent of σ, τ, p, q , and f .

Proof. By Lemma 5.7,

$$(5.41) \quad \left\| \{\lambda_{\sigma, I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)},$$

uniformly in $\sigma \in G$.

Fix

$$\sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N.$$

For $\gamma \in G$, define

$$U_\gamma : L^2(\Gamma, dw) \longrightarrow L^2(\gamma\Gamma, dw)$$

by

$$U_\gamma u(x) := u(\gamma^{-1}x), \quad x \in \gamma\Gamma.$$

Since dw is G -invariant, U_γ is unitary. In fact, the same argument shows that U_γ is an isometry on $L^s(dw)$ for every $1 \leq s \leq \infty$.

For $I \in \mathbb{D}_\Gamma$, define

$$\widetilde{e}_I := U_\sigma^{-1} e_{p, q, I}^{\sigma, \tau}$$

and

$$\widetilde{h}_I := U_\tau^{-1} \overline{h_{p,q,I}^{\sigma,\tau}}.$$

By (??),

$$\text{supp } \widetilde{e}_I \subset I, \quad \text{supp } \widetilde{h}_I \subset I^*.$$

We claim that

$$\{\widetilde{e}_I\}_{I \in \mathbb{D}_\Gamma} \quad \text{and} \quad \{\widetilde{h}_I\}_{I \in \mathbb{D}_\Gamma}$$

are NWO sequences.

Indeed, write

$$I = \Phi^{-1}(Q), \quad I^* = \Phi^{-1}((2Q) \cap \Omega).$$

Since Φ and Φ^{-1} are bi-Lipschitz, the set I^* is contained in a fixed geometric enlargement of I , with the enlargement constant independent of I . Thus the support condition in the definition of an NWO sequence is satisfied for both families.

Moreover, since U_σ and U_τ preserve the $L^r(dw)$ norm, (??) gives

$$(5.42) \quad \|\widetilde{e}_I\|_{L^r(\Gamma, dw)} = \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}$$

and

$$(5.43) \quad \|\widetilde{h}_I\|_{L^r(\Gamma, dw)} = \|h_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

Since $r > 2$, these are precisely the required L^r normalization conditions for NWO sequences. Therefore

$$\{\widetilde{e}_I\}_{I \in \mathbb{D}_\Gamma} \quad \text{and} \quad \{\widetilde{h}_I\}_{I \in \mathbb{D}_\Gamma}$$

are NWO sequences.

Consider now the operator on $L^2(\Gamma, dw)$ given by

$$(5.44) \quad \widetilde{A}_{\sigma,\tau,p,q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \left\langle \cdot, \widetilde{h}_I \right\rangle_{L^2(\Gamma, dw)} \widetilde{e}_I.$$

Applying Lemma 2.8 with exponent N , and then using (5.41), we obtain

$$(5.45) \quad \|\widetilde{A}_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\Gamma, dw))} \lesssim \|\{\lambda_{\sigma,I}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{N,\infty}} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

It remains to transfer this estimate back to $L^2(\mathbb{R}^N, dw)$.

By (??), the range of $A_{\sigma,\tau,p,q}(f)$ is contained in $L^2(\sigma\Gamma, dw)$, whereas the operator vanishes on $L^2(\mathbb{R}^N \setminus \tau\Gamma, dw)$. Thus its nonzero part is an operator

$$L^2(\tau\Gamma, dw) \longrightarrow L^2(\sigma\Gamma, dw).$$

For $g \in L^2(\tau\Gamma, dw)$, put

$$u := U_\tau^{-1}g.$$

Then, using the definition of $\widetilde{h}_I$ and the unitarity of U_τ ,

$$\left\langle g, \overline{h_{p,q,I}^{\sigma,\tau}} \right\rangle_{L^2(\tau\Gamma, dw)} = \left\langle U_\tau^{-1}g, U_\tau^{-1} \overline{h_{p,q,I}^{\sigma,\tau}} \right\rangle_{L^2(\Gamma, dw)} = \langle u, \widetilde{h}_I \rangle_{L^2(\Gamma, dw)}.$$

Also,

$$e_{p,q,I}^{\sigma,\tau} = U_\sigma \widetilde{e}_I.$$

Consequently,

$$\begin{aligned} A_{\sigma,\tau,p,q}(f)g &= \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle g, \overline{h_{p,q,I}^{\sigma,\tau}} \rangle e_{p,q,I}^{\sigma,\tau} \\ &= U_\sigma \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle u, \widetilde{h}_I \rangle \widetilde{e}_I \\ &= U_\sigma \widetilde{A}_{\sigma,\tau,p,q}(f) U_\tau^{-1} g. \end{aligned}$$

Hence

$$(5.46) \quad A_{\sigma,\tau,p,q}(f) = U_\sigma \widetilde{A}_{\sigma,\tau,p,q}(f) U_\tau^{-1}$$

on $L^2(\tau\Gamma, dw)$, with the left-hand side extended by zero on the orthogonal complement of $L^2(\tau\Gamma, dw)$.

Therefore the nonzero singular values of $A_{\sigma,\tau,p,q}(f)$ and $\widetilde{A}_{\sigma,\tau,p,q}(f)$ coincide. In particular,

$$\|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} = \|\widetilde{A}_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\Gamma, dw))}.$$

Combining this identity with (5.45), we conclude that

$$\|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

The implicit constant is independent of σ, τ, p, q , and f , which proves (5.40). $\square$

Lemma 5.9. *Assume that $N > 2$. Then there exists a constant $C_{N,R,k} > 0$ such that, for every $f \in L^N(\mathbb{R}^N, dx)$,*

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} \in S^{N,\infty}(L^2(\mathbb{R}^N, dw))$$

and

$$(5.47) \quad \|M_f(-\Delta_{\text{Dunkl}})^{-1/2}\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} \leq C_{N,R,k} \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. Choose $2 < r < N$. By Lemma 5.7, $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$, and hence Lemma 5.4 is applicable with this choice of r .

Let

$$\{c_{p,q}\}_{p,q \in \mathbb{Z}^N}, \quad \{e_{p,q,I}^{\sigma,\tau}\}, \quad \{h_{p,q,I}^{\sigma,\tau}\}$$

be the coefficients and functions given by Lemma 5.4. For $\sigma, \tau \in G$ and $p, q \in \mathbb{Z}^N$, let

$$A_{\sigma,\tau,p,q}(f) := \sum_{I \in \mathbb{D}_\Gamma} \lambda_{\sigma,I}(f) \langle \cdot, \overline{h_{p,q,I}^{\sigma,\tau}} \rangle_{L^2(\mathbb{R}^N, dw)} e_{p,q,I}^{\sigma,\tau}.$$

By Lemma 5.8,

$$(5.48) \quad \|A_{\sigma,\tau,p,q}(f)\|_{S^{N,\infty}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{L^N(\mathbb{R}^N, dx)}$$

uniformly in

$$\sigma, \tau \in G, \quad p, q \in \mathbb{Z}^N.$$

On the other hand, by (??), for every $M > 0$,

$$c_{p,q} \leq C_M (1 + |p| + |q|)^{-M}.$$

Taking $M > 2N$, we obtain

$$(5.49) \quad \sum_{p,q \in \mathbb{Z}^N} c_{p,q} < \infty.$$

Since G is finite, (5.48) and (5.49) imply

$$\begin{aligned} & \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p, q} \|A_{\sigma, \tau, p, q}(f)\|_{S^{N, \infty}(L^2(\mathbb{R}^N, dw))} \\ & \lesssim |G|^2 \left(\sum_{p, q \in \mathbb{Z}^N} c_{p, q} \right) \|f\|_{L^N(\mathbb{R}^N, dx)} \\ & \lesssim_{N, R, k} \|f\|_{L^N(\mathbb{R}^N, dx)}. \end{aligned}$$

Therefore the series

$$\sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p, q} A_{\sigma, \tau, p, q}(f)$$

converges in $S^{N, \infty}(L^2(\mathbb{R}^N, dw))$, and

$$(5.50) \quad \left\| \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p, q} A_{\sigma, \tau, p, q}(f) \right\|_{S^{N, \infty}(L^2(\mathbb{R}^N, dw))} \lesssim_{N, R, k} \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

It remains only to identify the operator represented by this series. By the definition of $A_{\sigma, \tau, p, q}(f)$, the integral kernel of its summand corresponding to $I \in \mathbb{D}_\Gamma$ is

$$\lambda_{\sigma, I}(f) e_{p, q, I}^{\sigma, \tau}(x) h_{p, q, I}^{\sigma, \tau}(y).$$

Hence the kernel identity (5.32) in Lemma 5.4 gives

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} = C \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p, q} A_{\sigma, \tau, p, q}(f).$$

More precisely, the two sides have the same integral kernel

$$f(x) K_{(-\Delta_{\text{Dunkl}})^{-1/2}}(x, y)$$

for almost every (x, y) with $d(x, y) > 0$; hence they agree initially on the natural test-function class and therefore, by the boundedness of the right-hand side, as operators on $L^2(\mathbb{R}^N, dw)$.

Combining this identity with (5.50), we obtain

$$M_f(-\Delta_{\text{Dunkl}})^{-1/2} \in S^{N, \infty}(L^2(\mathbb{R}^N, dw))$$

and

$$\|M_f(-\Delta_{\text{Dunkl}})^{-1/2}\|_{S^{N, \infty}(L^2(\mathbb{R}^N, dw))} \lesssim_{N, R, k} \|f\|_{L^N(\mathbb{R}^N, dx)}.$$

Absorbing the structural constants into $C_{N, R, k}$ gives (5.47). $\square$

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